

1. Compute the matrices $M(\sigma)$ for all $\sigma \in \mathfrak{S}_3$ in the natural representation. Compute the traces of these matrices, and see if there is some pattern in the traces.

Solution:

Recall that in the natural representation of $\mathfrak{S}_3$ on $\mathbb{C}[3] = \mathbb{C}\{\underline{1}, \underline{2}, \underline{3}\}$, the action is $\sigma \cdot \underline{i} = \underline{\sigma(i)}$. In the basis $(\underline{1}, \underline{2}, \underline{3})$, the matrix $M(\sigma)$ is the permutation matrix whose j -th column is $e_{\sigma(j)}$, i.e. the entry $(M(\sigma))_{i,j} = \delta_{\sigma(j),i}$.

Listing all six elements:

$$M(e) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad M((12)) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad M((13)) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

$$M((23)) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad M((123)) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad M((132)) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

The traces are

$$\text{Tr}M(e) = 3, \quad \text{Tr}M((12)) = \text{Tr}M((13)) = \text{Tr}M((23)) = 1, \quad \text{Tr}M((123)) = \text{Tr}M((132)) = 0.$$

Pattern. For each σ , $\text{Tr}M(\sigma) = |\text{Fix}(\sigma)| = \#\{i \in [3] : \sigma(i) = i\}$. Indeed, the diagonal entry $(M(\sigma))_{i,i} = \delta_{\sigma(i),i}$ equals 1 exactly when i is fixed by σ . Moreover, the trace is constant on conjugacy classes: it depends only on the cycle type of σ (the identity has 3 fixed points, transpositions have 1, and 3-cycles have 0).

2. Show that the sign representation $\text{sgn}: \mathfrak{S}_n \rightarrow \mathbb{C}^*$ is not a permutation representation.

Solution:

Suppose, towards a contradiction, that the sign representation is a permutation representation. By definition, this means there is a finite set X on which $\mathfrak{S}_n$ acts and a $\mathfrak{S}_n$ -equivariant linear isomorphism $\varphi: \mathbb{C}X \rightarrow \mathbb{C}$. In particular, $\dim \mathbb{C}X = \dim \mathbb{C} = 1$, so $|X| = 1$, say $X = \{x_0\}$. Since the only function from $\mathfrak{S}_n$ to permutations of a one-point set is trivial, we must have $\sigma \cdot x_0 = x_0$ for all $\sigma \in \mathfrak{S}_n$. Hence the action of $\mathfrak{S}_n$ on $\mathbb{C}X$ is the trivial representation, and so sgn would have to equal the trivial character, i.e. $\text{sgn}(\sigma) = 1$ for all σ .

But for $n \geq 2$, $\text{sgn}((12)) = -1 \neq 1$, a contradiction. Therefore sgn is not a permutation representation.

(Equivalent formulation: in a permutation representation the matrix of every group element is a permutation matrix, in particular has nonnegative real entries. For

$n \geq 2$, the 1×1 “matrix” of (12) in the sign representation is $[-1]$, which is not a permutation matrix.)

3. Show that if $H_1 = \langle(12)\rangle$, $H_2 = \langle(13)\rangle$, $H_3 = \langle(23)\rangle$, then the representations on $\mathfrak{S}_3/H_1$, $\mathfrak{S}_3/H_2$, and $\mathfrak{S}_3/H_3$ are all isomorphic.

Solution:

We use the fact: if $H, K \leq G$ $K = gHg^{-1}$ for some $g \in G$, if and only if the coset G -sets G/H and G/K are isomorphic.

After linearization, this G -set isomorphism induces an isomorphism of permutation representations $\mathbb{C}\{G/H\} \cong_G \mathbb{C}\{G/K\}$.

It remains to show that H_1, H_2, H_3 are pairwise conjugate in $\mathfrak{S}_3$. This follows because all transpositions in $\mathfrak{S}_n$ form a single conjugacy class. Concretely:

$$(13)(12)(13)^{-1} = (32) = (23), \quad (23)(12)(23)^{-1} = (13).$$

Hence $(13)H_1(13)^{-1} = H_3$ and $(23)H_1(23)^{-1} = H_2$. By the general fact above,

$$\mathfrak{S}_3/H_1 \cong_{\mathfrak{S}_3} \mathfrak{S}_3/H_2 \cong_{\mathfrak{S}_3} \mathfrak{S}_3/H_3,$$

and the same isomorphisms persist after passing to the corresponding permutation representations.

4. Find a subgroup $H \leq \mathfrak{S}_n$ such that the representation on $\mathfrak{S}_n/H$ is isomorphic to the natural representation of $\mathfrak{S}_n$.

Solution:

Take $H = \text{St}_{\mathfrak{S}_n}(\underline{n}) = \{\sigma \in \mathfrak{S}_n : \sigma(n) = n\} \cong \mathfrak{S}_{n-1}$, the stabilizer of the point n in the natural action of $\mathfrak{S}_n$ on $[n] = \{1, 2, \dots, n\}$.

The natural action of $\mathfrak{S}_n$ on $[n]$ is transitive, so by the orbit–stabilizer theorem, the map

$$\Psi: \mathfrak{S}_n/H \longrightarrow [n], \quad \sigma H \mapsto \sigma(n)$$

is a well-defined $\mathfrak{S}_n$ -equivariant bijection. Indeed, if $\sigma H = \sigma' H$ then $\sigma' = \sigma h$ for some $h \in H$, and $\sigma'(n) = \sigma(h(n)) = \sigma(n)$; equivariance is immediate from $(\tau\sigma)(n) = \tau(\sigma(n))$; surjectivity is the transitivity of the action; and the two finite sets have the same cardinality n .

Linearizing Ψ gives a $\mathfrak{S}_n$ -isomorphism between the coset representation $\mathbb{C}\{\mathfrak{S}_n/H\}$ and the natural representation $\mathbb{C}[n]$.

5. Show that the regular representation of $\mathfrak{S}_3$ is *not* isomorphic to the conjugation representation of $\mathfrak{S}_3$ (where G acts on $\mathbb{C}G$ by $g \cdot h = ghg^{-1}$).

Solution:

First solution using a G -isomorphism. Suppose there exists a linear isomorphism $\phi : \mathbb{C}\mathfrak{S}_3^{\text{conj}} \rightarrow \mathbb{C}\mathfrak{S}_3^{\text{reg}}$ such that for all $\sigma, \tau \in \mathfrak{S}_3$,

$$\phi(\sigma \cdot \tau) = \sigma \cdot \phi(\tau).$$

According to the actions on the two representations, this means that for all $\sigma, \tau \in \mathfrak{S}_3$ we must have

$$\phi(\sigma\tau\sigma^{-1}) = \sigma\phi(\tau).$$

If we choose an element $x \in \mathbb{C}\mathfrak{S}_3^{\text{conj}}$ such that for all $\sigma \in \mathfrak{S}_3$,

$$x = \sigma x \sigma^{-1}, \tag{1}$$

then we must have

$$\phi(x) = \phi(\sigma x \sigma^{-1}) = \sigma \phi(x).$$

This implies that if

$$\phi(x) = a_e e + a_{(12)}(12) + a_{(13)}(13) + a_{(23)}(23) + a_{(123)}(123) + a_{(132)}(132),$$

then $a_e = a_{(12)} = a_{(13)} = a_{(23)} = a_{(123)} = a_{(132)}$, since the regular action on $\mathfrak{S}_3$ has a unique orbit (it is transitive). We would then have

$$\phi(x) = c_x (e + (12) + (13) + (23) + (123) + (132)) = c_x \sum_{\tau \in \mathfrak{S}_3} \tau.$$

Now note that the elements $e, (123) + (132), (12) + (13) + (23) \in \mathbb{C}\mathfrak{S}_3^{\text{conj}}$ all satisfy equation (1), since they are sums over the elements of the 3 different orbits of the conjugation action on $\mathfrak{S}_3$.

By the discussion above, if $\phi(e) = c_e \sum_{\tau \in \mathfrak{S}_3} \tau$ and $\phi((123) + (132)) = c_{(123)+(132)} \sum_{\tau \in \mathfrak{S}_3} \tau$, then

$$\phi(c_{(123)+(132)} e - c_e((123) + (132))) = 0.$$

This means there is a nonzero element $c_{(123)+(132)} e - c_e((123) + (132)) \neq 0$ in $\ker \phi$, so ϕ cannot be an isomorphism, a contradiction.

Another solution using characters. We compute the characters of both representations and observe that they differ. Since isomorphic representations have equal characters, this suffices.

Both representations have $G = \mathfrak{S}_3$ acting on the vector space $\mathbb{C}G$, which has basis G itself. In each case, the basis is permuted by the action, so each is a permutation

representation, and Problem 9 below tells us that the character at g counts fixed basis elements.

Regular representation. For $g \in G$, the fixed elements of the basis are $\{h \in G : gh = h\} = \{h : g = e\}$. Hence

$$\chi^{\text{reg}}(g) = \begin{cases} |G| = 6 & \text{if } g = e, \\ 0 & \text{otherwise.} \end{cases}$$

On the three conjugacy classes $\{e\}, \{(12), (13), (23)\}, \{(123), (132)\}$, we get $\chi^{\text{reg}} = (6, 0, 0)$.

Conjugation representation. For $g \in G$, the fixed basis elements are $\{h \in G : ghg^{-1} = h\} = C_G(g)$, the centralizer of g . So $\chi^{\text{conj}}(g) = |C_G(g)|$. For $\mathfrak{S}_3$ we have:

- $C_{\mathfrak{S}_3}(e) = \mathfrak{S}_3$, so $\chi^{\text{conj}}(e) = 6$;
- $C_{\mathfrak{S}_3}((12)) = \langle (12) \rangle$, so $\chi^{\text{conj}}((12)) = 2$ (similarly for $(13), (23)$);
- $C_{\mathfrak{S}_3}((123)) = \langle (123) \rangle$, so $\chi^{\text{conj}}((123)) = 3$ (similarly for (132)).

Thus on the three conjugacy classes, $\chi^{\text{conj}} = (6, 2, 3)$.

Since $\chi^{\text{reg}} \neq \chi^{\text{conj}}$ (e.g., $\chi^{\text{reg}}((12)) = 0 \neq 2 = \chi^{\text{conj}}((12))$), the two representations are not isomorphic.

6. Let M be an R -module and let N_1, N_2 be R -submodules. Show that $N_1 + N_2$ is an R -submodule of M , and that $N_1 \cap N_2$ is an R -submodule of M .

Solution:

Recall $N_1 + N_2 = \{n_1 + n_2 : n_1 \in N_1, n_2 \in N_2\} \subseteq M$.

$N_1 + N_2$ is a submodule.

- *Contains 0:* $0 = 0 + 0$ with $0 \in N_1$ and $0 \in N_2$.
- *Closed under addition:* For $n_1 + n_2, n'_1 + n'_2 \in N_1 + N_2$,

$$(n_1 + n_2) + (n'_1 + n'_2) = (n_1 + n'_1) + (n_2 + n'_2) \in N_1 + N_2,$$
since $n_1 + n'_1 \in N_1$ and $n_2 + n'_2 \in N_2$ by closure of each N_i .
- *Closed under scalar multiplication:* For $r \in R$ and $n_1 + n_2 \in N_1 + N_2$,

$$r(n_1 + n_2) = rn_1 + rn_2 \in N_1 + N_2,$$

since $rn_1 \in N_1$ and $rn_2 \in N_2$.

$N_1 \cap N_2$ is a submodule.

- *Contains 0:* $0 \in N_1$ and $0 \in N_2$, so $0 \in N_1 \cap N_2$.
- *Closed under addition:* If $x, y \in N_1 \cap N_2$, then $x, y \in N_1$ implies $x + y \in N_1$, and similarly $x + y \in N_2$, so $x + y \in N_1 \cap N_2$.
- *Closed under scalar multiplication:* If $r \in R$ and $x \in N_1 \cap N_2$, then $rx \in N_1$ and $rx \in N_2$, hence $rx \in N_1 \cap N_2$.

This completes the proof.

7. Show that the standard representation $V_2 = \mathbb{C}\{\underline{2} - \underline{1}, \underline{3} - \underline{1}\}$ of $\mathfrak{S}_3$ is irreducible.

Hint: If $W \subseteq V_2$ is a nonzero $\mathfrak{S}_3$ -submodule, show $\dim W = 2$ by considering the action of generators on a nonzero vector in W .

Solution:

Let $e_1 := \underline{2} - \underline{1}$ and $e_2 := \underline{3} - \underline{1}$, so $V_2 = \mathbb{C}\{e_1, e_2\}$ has dimension 2. Suppose $W \subseteq V_2$ is a nonzero $\mathfrak{S}_3$ -submodule. Since $\dim W \in \{1, 2\}$ and $W \subseteq V_2$, it suffices to rule out $\dim W = 1$.

Suppose, for contradiction, that $W = \mathbb{C}w$ for some nonzero $w = \alpha e_1 + \beta e_2$. Then for every $\sigma \in \mathfrak{S}_3$, the vector $\sigma \cdot w$ must be a scalar multiple of w . We will get a contradiction using the generators (12) and (23).

Action of (12). Since (12) swaps $\underline{1} \leftrightarrow \underline{2}$ and fixes $\underline{3}$,

$$\begin{aligned} (12) \cdot e_1 &= (12)(\underline{2} - \underline{1}) = \underline{1} - \underline{2} = -e_1, \\ (12) \cdot e_2 &= (12)(\underline{3} - \underline{1}) = \underline{3} - \underline{2} = (\underline{3} - \underline{1}) - (\underline{2} - \underline{1}) = -e_1 + e_2. \end{aligned}$$

Hence $(12) \cdot w = -\alpha e_1 + \beta(-e_1 + e_2) = -(\alpha + \beta)e_1 + \beta e_2$. The condition $(12) \cdot w \in \mathbb{C}w$ requires this to be a scalar multiple $\lambda w = \lambda \alpha e_1 + \lambda \beta e_2$:

$$-(\alpha + \beta) = \lambda \alpha \quad \text{and} \quad \beta = \lambda \beta. \quad (*)$$

Action of (23). Since (23) swaps $\underline{2} \leftrightarrow \underline{3}$ and fixes $\underline{1}$,

$$\begin{aligned} (23) \cdot e_1 &= (23)(\underline{2} - \underline{1}) = \underline{3} - \underline{1} = e_2, \\ (23) \cdot e_2 &= (23)(\underline{3} - \underline{1}) = \underline{2} - \underline{1} = e_1. \end{aligned}$$

So $(23) \cdot w = \alpha e_2 + \beta e_1 = \beta e_1 + \alpha e_2$. The condition $(23) \cdot w \in \mathbb{C}w$ requires $(\beta, \alpha) = \mu(\alpha, \beta)$ for some $\mu \in \mathbb{C}$:

$$\beta = \mu \alpha \quad \text{and} \quad \alpha = \mu \beta. \quad (**)$$

From (**): $\alpha = \mu\beta = \mu(\mu\alpha) = \mu^2\alpha$, so either $\alpha = 0$ or $\mu^2 = 1$. Symmetrically, either $\beta = 0$ or $\mu^2 = 1$.

Case 1: $\mu^2 = 1$, i.e. $\mu = \pm 1$. Then $\beta = \mu\alpha = \pm\alpha$, so w is a scalar multiple of $e_1 + e_2 = \underline{2} + \underline{3} - \underline{2}\underline{1}$ or of $e_1 - e_2 = \underline{2} - \underline{3}$. Plug into (*):

- If $\beta = \alpha$ (so $\mu = 1$), then $-(2\alpha) = \lambda\alpha$ and $\alpha = \lambda\alpha$, forcing $\alpha = 0$ (else $\lambda = -2$ and $\lambda = 1$ contradict).
- If $\beta = -\alpha$ (so $\mu = -1$), then $0 = \lambda\alpha$ and $-\alpha = \lambda(-\alpha)$, forcing $\alpha = 0$ (else $\lambda = 0$ and $\lambda = 1$ contradict).

In both subcases $\alpha = \beta = 0$, contradicting $w \neq 0$.

Case 2: $\alpha = 0$. Then $w = \beta e_2$ with $\beta \neq 0$. From (**): $\beta = \mu \cdot 0 = 0$, contradicting $\beta \neq 0$.

Case 3: $\beta = 0$. Symmetric: $w = \alpha e_1$ with $\alpha \neq 0$, but (**) gives $\alpha = \mu \cdot 0 = 0$.

All cases lead to a contradiction, so no 1-dimensional $\mathfrak{S}_3$ -submodule of V_2 exists. Hence $W = V_2$, and V_2 is irreducible.

8. Let $\mathbb{Z}_n = \langle g \rangle$ be the cyclic group of order n . Show that the n one-dimensional representations $\rho_q(g) = e^{2\pi i q/n}$ for $q = 0, 1, \dots, n-1$ are pairwise non-isomorphic.

Hint: Two one-dimensional representations are isomorphic if and only if they are equal as functions.

Solution:

By the hint, it is enough to verify that the functions $\rho_q: \mathbb{Z}_n \rightarrow \mathbb{C}^*$ are pairwise distinct for $q = 0, 1, \dots, n-1$. Since $\mathbb{Z}_n = \langle g \rangle$, a homomorphism is determined by its value on g , so it suffices to check $\rho_q(g) \neq \rho_p(g)$ when $0 \leq p < q \leq n-1$.

Suppose $\rho_q(g) = \rho_p(g)$, i.e. $e^{2\pi i q/n} = e^{2\pi i p/n}$. Then

$$e^{2\pi i(q-p)/n} = 1,$$

which forces $\frac{q-p}{n} \in \mathbb{Z}$, i.e. $n \mid (q-p)$. But $0 \leq q-p \leq n-1$, so the only possibility is $q-p = 0$, i.e. $q = p$. Therefore the maps $\rho_0, \rho_1, \dots, \rho_{n-1}$ are pairwise distinct as functions, hence pairwise non-isomorphic as representations.

9. Prove that if $V = \mathbb{C}X$ is a permutation representation of G on a finite set X , then $\chi^V(g) = |\{x \in X \mid g \cdot x = x\}|$ for all $g \in G$.

Hint: How are the matrices $M(g)$ defined in the permutation representation? What is the trace of such a matrix?

Solution:

Order $X = \{x_1, x_2, \dots, x_n\}$ and use this as a basis for $V = \mathbb{C}X$. By definition of the permutation representation, $g \cdot x_j = x_{\sigma(j)}$ where $\sigma \in \mathfrak{S}_X$ is the permutation of X induced by g . Therefore the matrix $M(g)$ in this basis has entries

$$(M(g))_{i,j} = \delta_{i,\sigma(j)} = \begin{cases} 1 & \text{if } g \cdot x_j = x_i, \\ 0 & \text{otherwise.} \end{cases}$$

In particular, the diagonal entry $(M(g))_{i,i} = \delta_{i,\sigma(i)}$ equals 1 exactly when $g \cdot x_i = x_i$, and 0 otherwise. Summing the diagonal:

$$\chi^V(g) = \text{Tr} M(g) = \sum_{i=1}^n (M(g))_{i,i} = \sum_{i=1}^n \mathbf{1}[g \cdot x_i = x_i] = |\{x \in X : g \cdot x = x\}|.$$

10. Let V be a G -module with $\dim V = n$. Show:

(a) $\chi^V(e) = n$.

(b) If g and h are conjugate in G (i.e. $h = aga^{-1}$ for some $a \in G$), then $\chi^V(g) = \chi^V(h)$.

Hint for (b): trace is conjugation-invariant.

Solution:

(a) Since V is a G -module, the action of the identity $e \in G$ is the identity transformation $\text{id}_V \in \text{GL}(V)$. In any basis of V , the matrix of id_V is the $n \times n$ identity matrix I_n , and so

$$\chi^V(e) = \text{Tr} I_n = n.$$

(b) Fix a basis of V , and write $M(g)$ for the matrix of the action of g . Since V is a representation, $M: G \rightarrow \text{GL}_n(\mathbb{C})$ is a homomorphism, so

$$M(h) = M(aga^{-1}) = M(a) M(g) M(a)^{-1}.$$

Trace is invariant under conjugation in $\text{GL}_n(\mathbb{C})$ (because $\text{Tr}(AB) = \text{Tr}(BA)$ for all square matrices, hence $\text{Tr}(PXP^{-1}) = \text{Tr}(P^{-1}PX) = \text{Tr} X$). Therefore

$$\chi^V(h) = \text{Tr} M(h) = \text{Tr}(M(a)M(g)M(a)^{-1}) = \text{Tr} M(g) = \chi^V(g).$$

11. Construct the character tables of the following groups:

- (a) $\mathbb{Z}_4$ (four irreducible characters, use $i = e^{2\pi i/4}$).
 (b) $\mathbb{Z}_2 \times \mathbb{Z}_2$ (four conjugacy classes, four one-dimensional irreducibles).

Solution:

Both groups are abelian of order 4, so each element forms its own conjugacy class, giving 4 classes and hence 4 irreducible representations. For abelian groups all irreducible representations are 1-dimensional, so the irreducible characters are exactly the group homomorphisms $G \rightarrow \mathbb{C}^*$.

(a) $\mathbb{Z}_4 = \langle g \mid g^4 = e \rangle$. By Problem 8, the four irreducible characters are $\chi_q(g^k) = e^{2\pi i qk/4} = i^{qk}$ for $q = 0, 1, 2, 3$.

$\mathbb{Z}_4$	e	g	g^2	g^3
χ_0	1	1	1	1
χ_1	1	i	-1	$-i$
χ_2	1	-1	1	-1
χ_3	1	$-i$	-1	i

(b) $\mathbb{Z}_2 \times \mathbb{Z}_2 = \langle a, b \mid a^2 = b^2 = e, ab = ba \rangle$. Any homomorphism $\chi: \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow \mathbb{C}^*$ is determined by $\chi(a), \chi(b) \in \{\pm 1\}$. The four characters are $\chi_{\varepsilon, \delta}(a^i b^j) = \varepsilon^i \delta^j$ for $\varepsilon, \delta \in \{\pm 1\}$.

$\mathbb{Z}_2 \times \mathbb{Z}_2$	e	a	b	ab
χ_{++}	1	1	1	1
χ_{-+}	1	-1	1	-1
χ_{+-}	1	1	-1	-1
χ_{--}	1	-1	-1	1

A quick orthogonality check confirms correctness: each row has norm-squared $\frac{1}{4}(1 + 1 + 1 + 1) = 1$, and any two distinct rows have inner product 0.

12. Let G be a finite group and let $\mathcal{F}_G = \{f: G \rightarrow \mathbb{C}\}$ be the $\mathbb{C}$ -vector space of all functions from G to $\mathbb{C}$. Define

$$\langle \alpha, \beta \rangle = \frac{1}{|G|} \sum_{g \in G} \alpha(g) \overline{\beta(g)}.$$

Show that $\langle \cdot, \cdot \rangle$ is an inner product on $\mathcal{F}_G$. That is, verify:

- (a) **Conjugate symmetry:** $\langle \alpha, \beta \rangle = \overline{\langle \beta, \alpha \rangle}$ for all $\alpha, \beta \in \mathcal{F}_G$.
 (b) **Linearity in the first argument:** $\langle c_1 \alpha_1 + c_2 \alpha_2, \beta \rangle = c_1 \langle \alpha_1, \beta \rangle + c_2 \langle \alpha_2, \beta \rangle$ for all $c_1, c_2 \in \mathbb{C}$ and $\alpha_1, \alpha_2, \beta \in \mathcal{F}_G$.

(c) **Positive definiteness:** $\langle \alpha, \alpha \rangle \geq 0$ for all α , with equality if and only if $\alpha = 0$.

Solution:

We use repeatedly the basic facts $\overline{z + w} = \bar{z} + \bar{w}$, $\overline{z\bar{w}} = \bar{z}w$, $\overline{\bar{z}} = z$, and $z\bar{z} = |z|^2 \geq 0$ for $z \in \mathbb{C}$.

(a) **Conjugate symmetry.**

$$\overline{\langle \beta, \alpha \rangle} = \overline{\frac{1}{|G|} \sum_{g \in G} \beta(g) \overline{\alpha(g)}} = \frac{1}{|G|} \sum_{g \in G} \overline{\beta(g)} \cdot \overline{\overline{\alpha(g)}} = \frac{1}{|G|} \sum_{g \in G} \alpha(g) \overline{\beta(g)} = \langle \alpha, \beta \rangle.$$

(We used that $1/|G|$ is a positive real, hence equals its own conjugate.)

(b) **Linearity in the first argument.** For $c_1, c_2 \in \mathbb{C}$ and $\alpha_1, \alpha_2, \beta \in \mathcal{F}_G$,

$$\begin{aligned} \langle c_1 \alpha_1 + c_2 \alpha_2, \beta \rangle &= \frac{1}{|G|} \sum_{g \in G} (c_1 \alpha_1(g) + c_2 \alpha_2(g)) \overline{\beta(g)} \\ &= c_1 \cdot \frac{1}{|G|} \sum_{g \in G} \alpha_1(g) \overline{\beta(g)} + c_2 \cdot \frac{1}{|G|} \sum_{g \in G} \alpha_2(g) \overline{\beta(g)} \\ &= c_1 \langle \alpha_1, \beta \rangle + c_2 \langle \alpha_2, \beta \rangle. \end{aligned}$$

(c) **Positive definiteness.**

$$\langle \alpha, \alpha \rangle = \frac{1}{|G|} \sum_{g \in G} \alpha(g) \overline{\alpha(g)} = \frac{1}{|G|} \sum_{g \in G} |\alpha(g)|^2.$$

Each summand $|\alpha(g)|^2 \geq 0$, and the prefactor $1/|G| > 0$, so $\langle \alpha, \alpha \rangle \geq 0$. Moreover, since this is a sum of nonnegative terms,

$$\langle \alpha, \alpha \rangle = 0 \iff |\alpha(g)|^2 = 0 \text{ for every } g \in G \iff \alpha(g) = 0 \text{ for every } g \iff \alpha = 0.$$

13. Let $V = \mathbb{C}\{\underline{2} - \underline{1}, \underline{3} - \underline{1}\}$ be the standard representation of $\mathfrak{S}_3$. Its character χ^V takes values:

$$\chi^V(e) = 2, \quad \chi^V((12)) = \chi^V((13)) = \chi^V((23)) = 0, \quad \chi^V((123)) = \chi^V((132)) = -1.$$

Verify that V is irreducible.

Hint: Compute $\langle \chi^V, \chi^V \rangle$.

Solution:

Recall the irreducibility criterion: a representation V of a finite group G is irreducible if and only if $\langle \chi^V, \chi^V \rangle = 1$. We compute this inner product directly. Since χ^V is constant on each conjugacy class, we can group the sum by classes, with sizes $|\{e\}| = 1$, $|\{(12), (13), (23)\}| = 3$, $|\{(123), (132)\}| = 2$:

$$\langle \chi^V, \chi^V \rangle = \frac{1}{|\mathfrak{S}_3|} \sum_{\sigma \in \mathfrak{S}_3} |\chi^V(\sigma)|^2 = \frac{1}{6} (1 \cdot |2|^2 + 3 \cdot |0|^2 + 2 \cdot |-1|^2) = \frac{1}{6} (4 + 0 + 2) = 1.$$

Therefore V is irreducible.

14. Show that the trivial representation $\mathbf{1}_{\mathfrak{S}_3}$, the sign representation $\text{sgn}_{\mathfrak{S}_3}$, and the standard representation V_3 are the *only* irreducible representations of $\mathfrak{S}_3$ (up to isomorphism).

Hint: How many conjugacy classes are there in $\mathfrak{S}_3$?

Solution:

Recall the fundamental theorem: the number of irreducible representations of a finite group G (up to isomorphism) equals the number of conjugacy classes of G .

Conjugacy classes of $\mathfrak{S}_3$. Two elements of $\mathfrak{S}_n$ are conjugate iff they have the same cycle type. The cycle types in $\mathfrak{S}_3$ are: $\{1, 1, 1\}$ (the identity), $\{2, 1\}$ (transpositions), $\{3\}$ (3-cycles). So $\mathfrak{S}_3$ has exactly 3 conjugacy classes:

$$\{e\}, \quad \{(12), (13), (23)\}, \quad \{(123), (132)\}.$$

Hence there are exactly 3 irreducible representations of $\mathfrak{S}_3$ up to isomorphism.

Three irreducibles already found. We exhibit three pairwise non-isomorphic irreducibles:

- $\mathbf{1}_{\mathfrak{S}_3}$: trivial, $\dim = 1$, character $(1, 1, 1)$.
- $\text{sgn}_{\mathfrak{S}_3}$: sign, $\dim = 1$, character $(1, -1, 1)$.
- V_3 : standard, $\dim = 2$, character $(2, 0, -1)$ (irreducible by Problem 13).

The first two are irreducible because they are 1-dimensional. They are pairwise non-isomorphic because their characters differ: $\mathbf{1}$ and sgn disagree at (12) , and V_3 has different dimension ($\chi(e) = 2$) from the other two ($\chi(e) = 1$).

Conclusion. Three pairwise non-isomorphic irreducibles in a list of length 3 exhausts all irreducible representations of $\mathfrak{S}_3$.

15. A function $f: G \rightarrow \mathbb{C}$ is a *class function* if $f(hgh^{-1}) = f(g)$ for all $g, h \in G$. Let $R(G)$ denote the set of all class functions on G . Show that $R(G)$ is a subspace of $\mathcal{F}_G$. That is, verify that $R(G)$ is closed under addition and scalar multiplication, and contains the zero function.

Solution:

We verify the three subspace axioms.

Zero function. The zero function $\mathbf{0}: G \rightarrow \mathbb{C}$, $\mathbf{0}(g) = 0$ for all g , satisfies $\mathbf{0}(hgh^{-1}) = 0 = \mathbf{0}(g)$ for all $g, h \in G$. Hence $\mathbf{0} \in R(G)$.

Closure under addition. Let $f_1, f_2 \in R(G)$ and $g, h \in G$. Then

$$(f_1 + f_2)(hgh^{-1}) = f_1(hgh^{-1}) + f_2(hgh^{-1}) = f_1(g) + f_2(g) = (f_1 + f_2)(g),$$

where the middle equality uses that each f_i is a class function. Hence $f_1 + f_2 \in R(G)$.

Closure under scalar multiplication. Let $f \in R(G)$, $c \in \mathbb{C}$, and $g, h \in G$. Then

$$(cf)(hgh^{-1}) = c \cdot f(hgh^{-1}) = c \cdot f(g) = (cf)(g),$$

so $cf \in R(G)$.

All three axioms hold, so $R(G)$ is a subspace of $\mathcal{F}_G$.

16. Use the character table of $\mathfrak{S}_4$ and Burnside's counting lemma to determine the number of distinct ways to color the faces of a cube with k colors, where two colorings are considered the same if one can be obtained from the other by a rotation.

Hint: The rotation group of the cube is isomorphic to $\mathfrak{S}_4$.

Solution:

Let G be the rotation group of the cube. By the hint, $G \cong \mathfrak{S}_4$ via the action of G on the four *long diagonals* of the cube; in particular $|G| = 24$. Let

$$X = \{\text{colorings } c: \{6 \text{ faces}\} \rightarrow \{1, \dots, k\}\}, \quad |X| = k^6,$$

on which G acts by $(g \cdot c)(f) = c(g^{-1}f)$. Two colorings are equivalent iff they lie in the same G -orbit, so the answer is $|X/G|$.

Step 1: Burnside via the character of the permutation representation.

Linearize the G -set X to get the permutation representation $V := \mathbb{C}X$ of G . By Maschke's theorem, V decomposes as a direct sum of irreducibles

$$V \cong_G \bigoplus_{\rho \in \text{Irr}(G)} m_\rho \cdot \rho, \quad m_\rho = \langle \chi^V, \chi^\rho \rangle.$$

The multiplicity of the trivial representation is, by Problem 9 and the orthogonality formula,

$$m_1 = \langle \chi^V, \chi^1 \rangle = \frac{1}{|G|} \sum_{g \in G} \chi^V(g) = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}_X(g)|.$$

On the other hand, the component $V^G \subseteq V$ formed by the copies of the trivial representation, has a basis indexed by the G -orbits on X (the orbit sums $\sum_{x \in \mathcal{O}} x$, one per orbit $\mathcal{O} \in X/G$, are linearly independent and span V^G). Hence

$$|X/G| = \dim V^G = m_1 = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}_X(g)|.$$

This is the character-theoretic incarnation of Burnside's counting lemma.

Step 2: The character table of $\mathfrak{S}_4$. Recall that $\mathfrak{S}_4$ has 5 conjugacy classes, and hence 5 irreducible representations. The character table is:

$\mathfrak{S}_4$ class size	e	(ab)	$(ab)(cd)$	(abc)	$(abcd)$
1	1	1	1	1	1
sgn	1	-1	1	1	-1
W (2-dim)	2	0	2	-1	0
V_3 (standard, 3-dim)	3	1	-1	0	-1
$\text{sgn} \otimes V_3$	3	-1	-1	0	1

The leftmost row gives the conjugacy class data we need: 5 classes of sizes 1, 6, 3, 8, 6, summing to $24 = |\mathfrak{S}_4|$.

Step 3: Compute χ^V on each class. Under $G \cong \mathfrak{S}_4$, each conjugacy class corresponds to a geometric rotation type. By Problem 9, $\chi^V(g) = |\text{Fix}_X(g)| = k^{c(g)}$, where $c(g)$ counts the cycles of g 's action on the 6 faces. We tabulate:

$\mathfrak{S}_4$ class	size	rotation type	cycles on 6 faces	χ^V
e	1	identity	1+1+1+1+1+1	k^6
(ab)	6	180° through edge midpoints	2+2+2	k^3
$(ab)(cd)$	3	180° through face centers	1+1+2+2	k^4
(abc)	8	±120° through vertex pair	3+3	k^2
$(abcd)$	6	±90° through face centers	1+1+4	k^3

(For the 90° face rotation: the two faces on the rotation axis are fixed and the four side faces form a 4-cycle; for the 180° face rotation: the same two are fixed and the

four side faces split into two transpositions; for the 120° vertex rotation: the three faces meeting at one corner are cycled, and so are the three at the opposite corner; for the 180° edge rotation: the six faces split into three transpositions.)

Step 4: Apply the formula. By Step 1,

$$\begin{aligned} |X/G| &= \langle \chi^V, \chi^1 \rangle = \frac{1}{24} \sum_{\text{classes}} (\text{size}) \cdot \chi^V \cdot \overline{\chi^1} = \frac{1}{24} (1 \cdot k^6 + 6 \cdot k^3 + 3 \cdot k^4 + 8 \cdot k^2 + 6 \cdot k^3) \\ &= \boxed{\frac{1}{24} (k^6 + 3k^4 + 12k^3 + 8k^2)}. \end{aligned}$$

Sanity checks.

- $k = 1$: $\frac{1}{24}(1 + 3 + 12 + 8) = \frac{24}{24} = 1$ (the unique monochromatic pattern).
- $k = 2$: $\frac{1}{24}(64 + 48 + 96 + 32) = \frac{240}{24} = 10$.
- $k = 3$: $\frac{1}{24}(729 + 243 + 324 + 72) = \frac{1368}{24} = 57$.

Alternative solution (Burnside directly). One can avoid the language of representations and apply Burnside's counting lemma straight to the G -set X :

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}_X(g)|.$$

A coloring $c \in X$ is fixed by $g \in G$ iff c is constant on each cycle of g acting on the 6 faces, so $|\text{Fix}_X(g)| = k^{c(g)}$. Using the same table of conjugacy class sizes and cycle counts as in Step 3 above, we obtain the same total

$$|X/G| = \frac{1}{24} (k^6 + 3k^4 + 12k^3 + 8k^2).$$

This is essentially the same calculation: in this approach the conjugacy class data of $\mathfrak{S}_4$ is used implicitly (as the partition of G over which the sum is taken), whereas in the main solution above it appears explicitly through the orthogonality relation $|X/G| = \langle \chi^V, \chi^1 \rangle$.